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Research Paper

A method of citrus epidermis defects detection based on an improved YOLOv5

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Achieving intelligent detection of citrus epidermal defects after harvesting is of great significance to citrus quality and value assurance. The uneven illumination makes it challenging to detect citrus epidermis abnormalities with great accuracy. In order to improve the detection accuracy of citrus epidermal invisible defects, firstly, a dual-lamp image acquisition system is designed and used to complete the image acquisition of citrus fruit invisible defects. Secondly, the YOLOv5 model was optimized by integrating the attention mechanism CBAM and modifying the loss function as DIoU. Finally, the performance of the improved model was verified by comparison experiments and ablation experiments. According to the experimental results, *mAP*, *Precision* and *Recall* of the improved YOLOv5 model were 95.5%, 94.0% and 95.1%, respectively, which were 5.8%, 3.6% and 7.6% higher than those of YOLOv5x. Meanwhile, the average detection speed increased by 22.1 ms per pic. This indicates that the improved network being applied to citrus epidermal defects detection can achieve better performance. It can provide technical support for the intelligent detection and grading of postharvest citrus.

1. Introduction

Citrus has a massive economic market, whether in China or the world (Chen et al., 2022). The quality classification of postharvest citrus plays an important role in further improving the competitiveness of the citrus market. One of the key tasks in the intelligent categorisation of postharvest citrus is the detection of epidermal defects (Lin, Tang, Zou, Cheng, 2020). In the intelligent detection of citrus quality, the colour and texture information of the defect area is highly similar to that of the normal area under natural light for most of the defects with not obvious features, such as damage and scars. The above invisible defects, often existing in citrus during the picking and transportation stages, can result in black and rotten epidermis over time, which will not only affect the appearance, but also deteriorate the internal quality, and even infect other good quality citrus, causing a certain degree of economic loss (Zhang et al., 2014). Therefore, it is of great significance to accurately detect the invisible defects of citrus epidermis.

Currently, studies on the detection of citrus epidermis defects mainly based on hyperspectral technologies and traditional machine vision methods. For example, López-García, Andreu-García, Blasco, and Aleixos (2010), Zhang, Lee, Li, and Zheng (2018), Tian, Fan, Huang, and Wang (2020),

Abbreviation: MLP, Multilayer Perceptron.

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Nomenclature

DIoU	Distance Intersection over Union
mAP	mean Average Precision
BottleneckCSP	Bottleneck Cross Stage Partial
GIoU	Generalised Intersection over Union
labelme	label by me
SiLU	Sigmoid-weighted linear units
FPN	Feature Pyramid Network
PAN	Path Aggregation Network
NMS	Non-Maximum Suppression
IoU	Intersection over Union
SENet	Squeeze and Excitation Network
CBAM	Convolutional Block Attention Module
ECANet	Efficient Channel Attention Network
CA	Coordinate Attention
CAM	Channel Attention Module
SAM	Spatial Attention Module
Relu	Rectified Linear Unit
CIoU	Complete Intersection over Union
EIoU	Efficient Intersection over Union

Zhang et al. (2021), Tian et al. (2021) and Luo et al. (2022) used hyperspectral imaging technology to obtain images of citrus infected parts, and realised the detection and classification of citrus epidermal defects based on the image processing algorithm. However, the cost of the above methods was not low, and their corresponding performance was poor.

Some scholars have conducted the identification of citrus epidermal invisible defects from the perspective of ultraviolet induced fluorescence imaging based on the biological properties of citrus under various excitation light sources. Kondo et al. (2009) found that fluorescence reaction in citrus epidermis could be excited by flavonoids with UV light. Blasco, Aleixos, Gómez and Moltó (2007) found that the essential oil of citrus epidermis defects produced yellow fluorescence in a dark room under UV light. The fluorescence reaction occurs owing to the high content of essential oils in citrus epidermis. As a component of essential oil in citrus peel, the Citrus Flavanone Naringenin (Polymethoxyflavone), infiltrated from epidermal cells, absorbs ultraviolet light energy to emit visible band fluorescence, a source of yellow fluorescence which can be visible from the damaged or decayed citrus epidermis in a dark room (Blasco et al., 2007). This is most likely due to damage caused by picking and transporting the fruit. Other defects, such as dark spots and scars on the peel, appear different from normal citrus areas due to higher levels of melanin (Chen, An, Gao, Li, 2021). This may be due to slight scarring (wind scarring) caused when the citrus fruit is blown against the branches while it is on the tree (Gómez-Sanchís et al., 2014). It is also possible that lesions may occur in the future due to spots caused by infections such as fungi (Lorente, Aleixos, Gómez-Sanchis, Cubero, 2013). The scars and spots on the skin will directly release grease, resulting in a very significant area under UV irradiation (Obenland, Margosan, Smilanick, 2010). The colour and texture information of the defect area under such natural light is highly similar to that in the normal area, but the defect that can be easily identified under ultraviolet light is called “invisible defect”. Siregar, Ahmad and Maddu (2018, May) detected citrus epidermis invisible defects based on machine vision system of ultraviolet induced fluorescence imaging. The above studies provide a new method for invisible defects detection in citrus epidermis with fluorescence image technology.

The fluorescence image technology is used to obtain citrus epidermis invisible defects images, also requires being combined with machine learning models for detection. With the development of machine learning, deep learning has been widely used in fruit detection (Lin, Tang, Zou, Li, 2019). Wan and Goudos (2020) proposed a multi-class fruit detection method using deep learning framework based on the improved Faster R–CNN with a mean Average Precision of over 91%. Chen, Lu, Liu, and Li (2020) realised rapid citrus recognition by improving the YOLOv4 detector, with a 3.15% increase of the mean Average Precision. Valdez (2020) conducted apple epidermal defects detection based on YOLOv3, *mAP*@0.5 was 74.3%; Kukreja and Dhiman (2020) conducted citrus defects detection based on the enhanced dataset of CNN, with an overall *Accuracy* of 89.1%. Dhiman et al. (2022) divided citrus fruit diseases into four grades and tested the severity of citrus diseases based on VGG16 and the results showed that this method could effectively detect the four severity degrees of citrus fruit diseases. Fan et al. (2022) proposed an apple defects detection method based on the improved YOLOv4 network, with a *mAP* of 93.74%. Tang, Zhou, Wang, and Zhang (2023) proposed a camellia fruit detection model based on the improved YOLOv4, with a *mAP* of 92.07%. Based on the above studies, it is feasible to detect citrus epidermal invisible defects based on deep learning.

In deep learning models, compared to the two-stage algorithm, the YOLO based one-stage detection algorithm discards the region extraction step and solves the detection task as a regression problem, which directly obtains the location and class information of the target in the images. Redmon, Divvala, Girshick, and Farhadi (2016) came up with the YOLO model. However, it has a relatively low recall rate and detection precision. Redmon & Farhadi (2017) proposed the YOLOv2 model. In comparison to the previous version, this model improves efficiency and detection speed. Redmon & Farhadi (2018) proposed YOLOv3. Bochkovskiy, Wang and Liao (2020) suggested YOLOv4, which has significantly improved model detection accuracy. Jocher, Stoken, Borovec and Christopher (2021) proposed the YOLOv5 model, a lightweight network based on the Pytorch framework. Compared with YOLOv4, YOLOv5 has faster inference speed and higher detection accuracy, and can achieve fast detection of 140 frames per second on Tesla P100. The minimum weight file size of YOLOv5 is only 2.62 MB, which makes the model simpler and easier to transfer to practical agricultural applications.

The YOLOv5 series is commonly used for target detection tasks, such as Kuznetsova, Maleva, Soloviev (2020), Kuznetsova, Maleva, Soloviev (2021), Wang, Jin, Wang, and Xu (2022), Yang, Hu, Yao, and Gao (2022), Wu et al. (2022) all used the algorithm to detect fruit targets, and good results have been obtained. Therefore, in this paper, we use the improved YOLOv5 for invisible defect detection in citrus epidermis.

In recent years, a number of scholars have proposed several improvements to the YOLOv5 algorithm. Yan, Fan, Lei, and Liu (2021) improved the YOLOv5s algorithm by replacing the BottleneckCSP module of the YOLOv5s backbone with BottleneckCSP-2 and adding the SE module, and by modifying the initial detection frame size of the YOLOv5s to the apple size, and the improved YOLOv5s *mAP* was 5.05% higher. Zhao et al. (2021) improved the network by adding a microscale detection layer and adjusting the loss function of the detection layer to achieve the detection of wheat ears on UAV images, and the improved network achieved an average precision at detection of 94.1%, which is 10.8% higher than that of the standard YOLOv5.

There is generally little research on the detection of epidermal invisible defects in citrus fruits, and there are issues with uneven citrus fruit illumination and challenging invisible defect identification in natural light. Therefore, this paper aims to propose a method for citrus epidermis defect detection based on an improved YOLOv5 with fluorescence imaging technique. First, a dual-lamp image acquisition system was designed and used to obtain images of citrus epidermis defects under UV and halogen lamps, respectively. Secondly, changing the loss function Generalised Intersection over Union (GIoU) to Distance Intersection over Union (DIoU) was to improve the problem of constant loss value when the real frame contains the predicted frame during convergence and accelerate convergence. The CBAM attention module was introduced into the backbone network of the YOLOv5 model, and all C3 structures were replaced by the C3CBAM module to improve the learning ability of the model for the target features, so as to achieve the improvement of the YOLOv5 model and used the improved model for invisible defects of citrus epidermis test. Finally, the improved model performance was verified by comparison and ablation experiments.

2. Materials and methods

2.1. Visual system

2.1.1. *Choice of light source*

It is important to use a light source that will work with the citrus epidermis' strong fluorescent reactions when using fluorescence image technology to identify invisible defects in citrus epidermis.

Momin et al. (2012) studied 15 varieties of citrus epidermal defects and the optimal wavelength of fluorescence excitation, observed the characteristics of UV absorbance, excitation and fluorescence spectrum, and classified citrus varieties according to fluorescence intensity level by principal component analysis and discriminant analysis. The results showed that the optimal fluorescence excitation wavelength and the resulting peak fluorescence wavelength varied with varieties, but the overall range was 350–380 nm and 490–540 nm, respectively. Kondo (2009) used a fluorescence spectrophotometer to test the spectrum of citrus skin extract in the study of citrus defects detection under UV light, and found that the wavelength of about 380 nm could excite the greatest fluorescence brightness. Based on the fluorescence reaction of citrus epidermal physical injury, Siregar, Ahmad, and Maddu (2018, July) found that the fluorescence excitation wavelength of local citrus was between 315, 340, 500 and 555 nm using fluorescence spectroscopy.

Therefore, a UV lamp with an emission wavelength of 350–380 nm and a peak value of 356 nm was selected as the illuminating light source for the experiment, and images of different citrus varieties were obtained and compared. It was found that not all citrus fruits had obvious fluorescence reactions. In combination with the experimental comparison of light source, the Zhejiang Linhai Citrus with the most obvious fluorescence phenomenon was selected as the experimental object, on which the following studies of citrus defects were based. According to the fluorescence spectrum data, Siregar et al. (2018, July) found that when the defective part of citrus was excited by 356 nm ultraviolet light, the fluorescence wavelength was 500–520 nm, that is, when the fluorescence wavelength was larger than the ultraviolet wavelength, the normal part of citrus epidermis and the defective part were different.

The fluorescence response of citrus defects was excited by ultraviolet light, and the white light illumination was realised by halogen lamps. Defects at the same position in the same citrus were compared under halogen and ultraviolet light. It can be seen from Fig. 1 that the invisible defects of citrus are not obvious under white light, but easy to identify under ultraviolet light. Meanwhile, the defects under ultraviolet light show a more accurate and complete area than those under white light. In addition, the citrus images captured under halogen lamp are with obvious shadows, while the shadows are unobvious under the ultraviolet lamp, which will be more favourable for subsequent image processing.

2.1.2. *Design of image acquisition system*

The visual system used in this study includes ultraviolet fluorescent lamps, halogen lamps, cameras, polarising film and filtering film, etc. Two 15 W UV lamps with peak excitation wavelength of 356 nm, length of 330 mm, and excitation angle of 45° diagonally downward, and two halogen lamps with a diameter of 35 mm, 220 V 50 W, and diagonally downward 60° were used as an illumination, shown as Fig. 2.

In order to obtain clearer images under pure ultraviolet light, namely the external light intensity should be less than 5lx, considering the easy usage and the parameters such as aperture, photosensitive element and so on, the iPhone12 and NIKON D5300 were selected as the cameras of the vision system in the dark room. The iPhone12 has a 12-megapixel model with an oversised wide-angle $f/2.4$ and a wide-angle $f/1.6$ aperture. The lens of the camera is Nikon AF-S DX NIKKOR, the capture mode is an automatic mode, and the ISO is 400.

The vision system fixed a polarising filter with a diameter of 52 mm in front of the camera to eliminate cross-polarised halos in the scene, as well as a 50 × 50 × 2 mm filter with a wavelength above 420 nm. Before image acquisition, some parameters need to be adjusted. When the UV lamp was used as the light source, the distance between the citrus centre and the camera was 270 mm, and the distance between the citrus centre and the ultraviolet lamp was 130 mm. When the halogen lamp was used as the light source, the distance between the centre of the citrus and the camera and halogen lamp was 220 and 190 mm respectively, shown as Fig. 2-b and

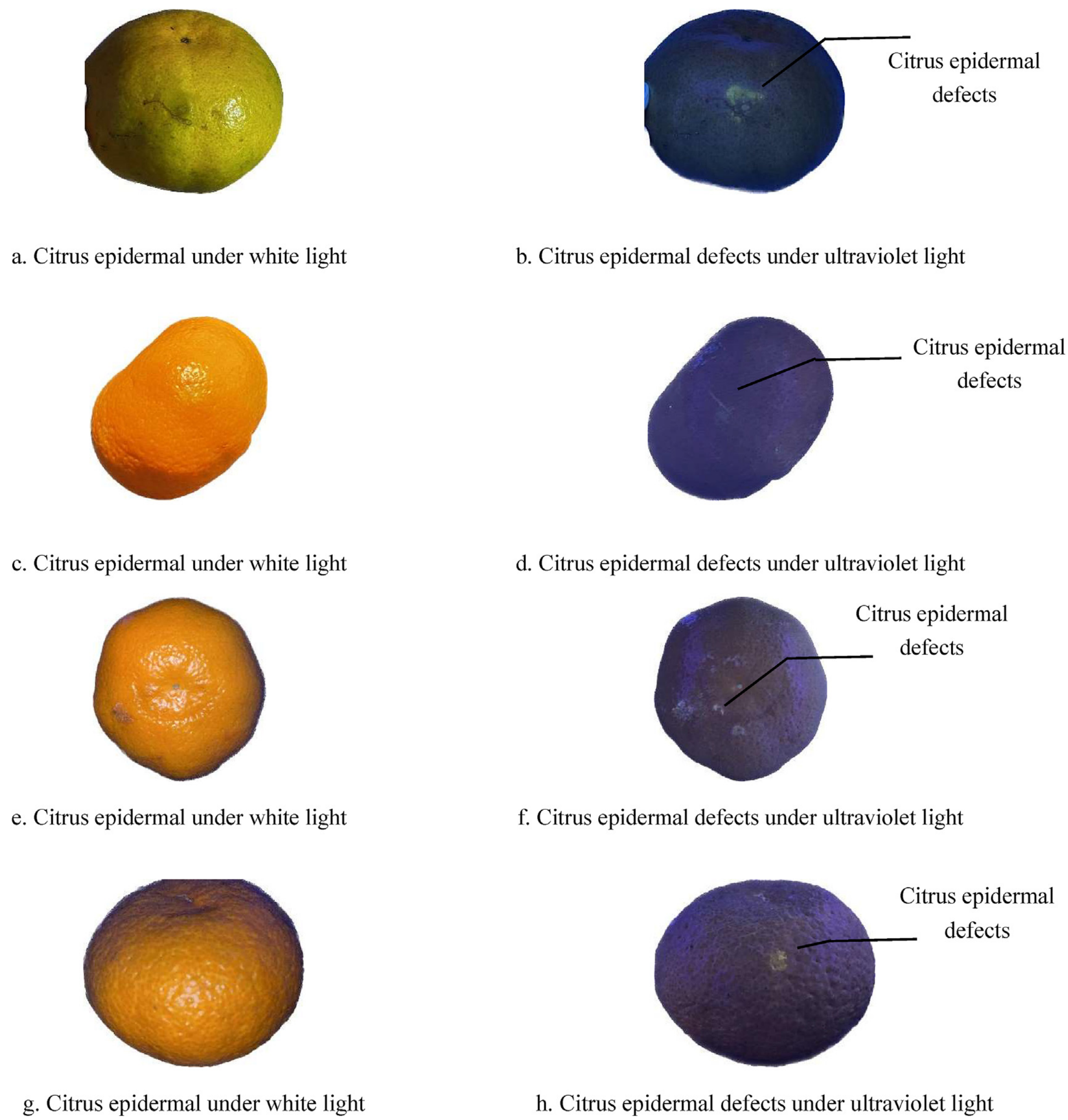


Fig. 1 – Images of citrus epidermal defects under different lights.

c. Based on these parameters, the image can be clear without distortion. Each sample was manually placed on the sample table and the fruit containing the defective area was positioned sideways towards the camera.

2.2. *Image sample acquisition*

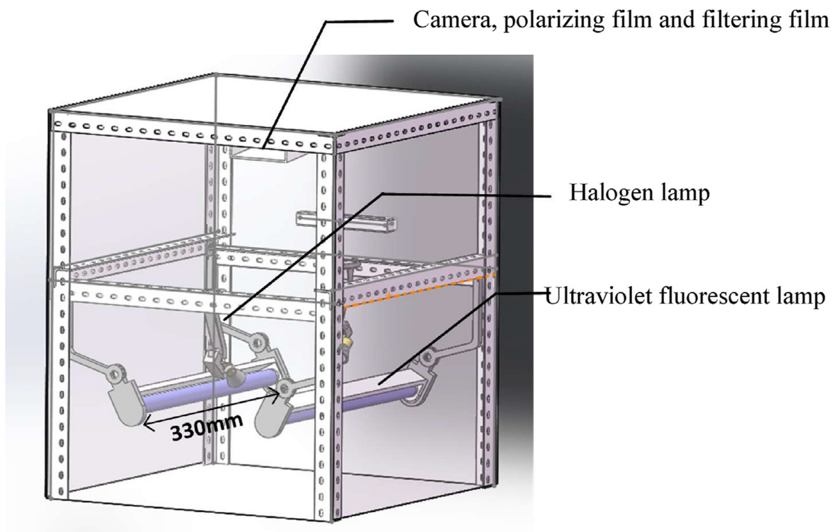
The citrus defects images were obtained in the laboratory of College of Mathematics and Information, South China Agricultural University, Guangzhou City, Guangdong Province, China. The image shooting object was Zhejiang Linhai Citrus, the acquisition time was from October to December 2021. The image resolution was 3024×4032 and 4496×3000 pixels, and the image format was JPG.

Images of defect areas of postharvest Zhejiang Linhai citrus were collected from different angles in the laboratory environment, and a total of 845 citrus epidermal defect images was obtained. The labelme tool was used to label the collected citrus epidermal defect images, and the annotated JSON files were converted to TXT coordinate labels in batches using python. Here, the label of defect type is defect1 for surface damage, and defect2 for black spots, as shown in Fig. 3. After random partition according to 6.5:1:1, there are 645 training sets, 100 verification sets and 100 test sets respectively. In order to improve the generalisation ability and robustness of the model, the training set was scaled, cropped and added noise, and there were 1524 images of the training set after image enhancement. The dataset distribution is shown in Table 1.

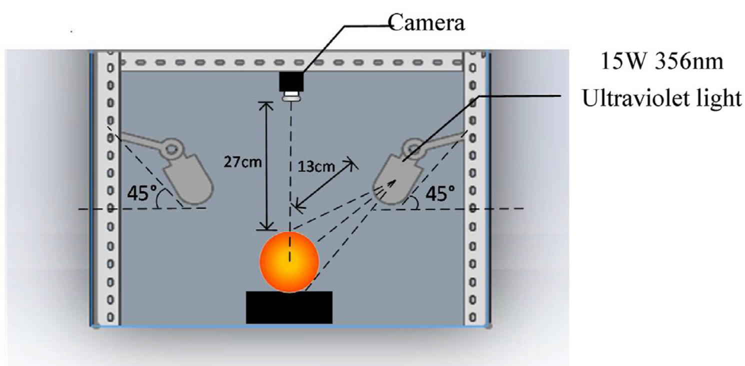
2.3. *YOLOv5 network and optimisation*

2.3.1. *YOLOv5 model*

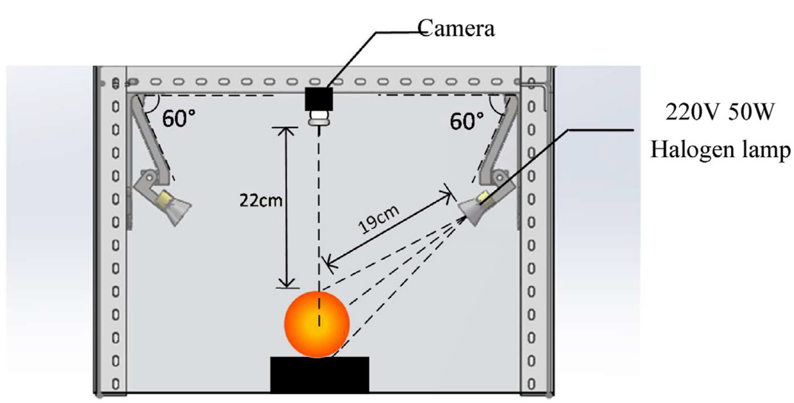
The YOLOv5 network model includes four components: Input, Backbone, Neck and Prediction. YOLOv5 adds adaptive anchor frame calculation, adaptive picture scaling, and Mosaic data augmentation to the Input module. By combining four photos and expanding the dataset through random scaling, random



a. Overall structure of the visual system



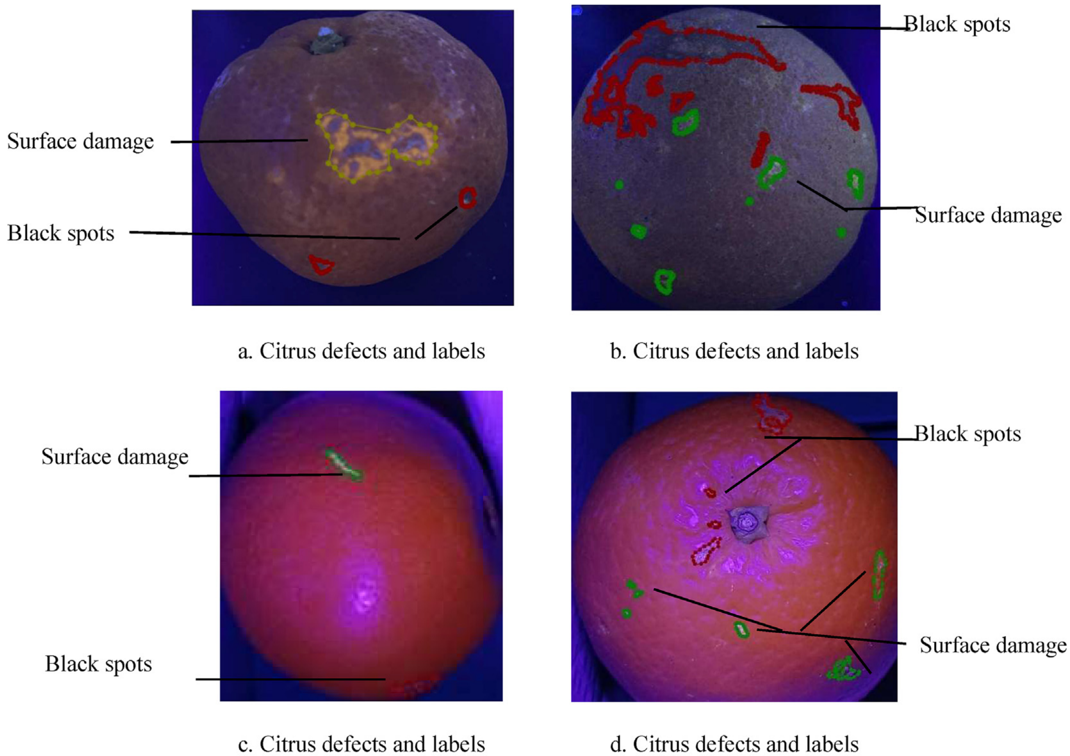
b. The image acquisition system under ultraviolet light



c. The image acquisition system under halogen lamps

Fig. 2 – Visual system structure diagram.

clipping, and random layout, mosaic data augmentation increases the network's robustness and enhances its capacity for small target recognition. When YOLOv3 and YOLOv4 are trained with different dataset, the initial anchor frame value is calculated by a separate program. However, YOLOv5 can decide the adaptive adjustment of anchor frame and image in each training process according to the actual situation. In this paper, images of different lengths and widths were used. YOLOv5 network uniformly changes the size of input images to 640 × 640, thus improving the speed of inference and detection.



Note: Yellow and green are marked as surface damage and red as black spots.

Fig. 3 – Examples of citrus defects and labels.

Table 1 – Distribution of dataset.

Type of defect	Defect1	Defect2	Defect1 & Defect2	The total number
Dataset	229	159	457	845
Set of training	155	99	391	645
Enhancement training set	310	297	917	1524
Set of verification	34	30	36	100
Set of tests	40	30	30	100

The Conv structure, C3 structure, and SPPF structure compose the main part of the Backbone module (Zhou, Zhao, Nie, 2021). Compared with version 5.0, YOLOv5 version 6.0 replaces Focus layer with Conv structure, which consists of the common convolution 2D layer, the batch normalisation layer BN and swish function (SiLU). With only three convolution modules in addition to its Bottleneck, the C3 module is a condensed version of the BottleneckCSP module. The Spatial Pyramid Pooling Fast (SPPF), as its name implies, uses three 5×5 maximum pools to replace the original 5×5, 9×9, and 13×13 maximum pools under the same kernel size in order to reduce the amount of computation and improve the speed under the same accuracy conditions. And the features are connected together for fusion.

The Neck module adopts Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) that is mainly applied to generate feature pyramids and enhance the model's capacity to detect objects of various scales.

The Prediction module includes the GIoU_Loss and the Non-Maximum Suppression (NMS). The NMS can erase unnecessary boxes and find the best object detection location.

2.3.2. *Loss function optimisation*

A basic task in computer vision is bounding box regression. So far, the loss function used commonly is the Intersection over Union (IoU) loss and its variants. GIoU was proposed based on the IoU loss function to solve the problem that IoU could not reflect how two boxes intersected. However, GIoU still has problems with the accuracy of adjacent defect recognition. For some target detection frames without overlap, GIoU regression strategy may degenerate into intersection ratio IoU.

For YOLOv5, the network loss was defined using rectangular box loss (box_loss), confidence loss (obj_loss), and classification loss (cls_loss) as formula (1):

$$Loss = a{*}box\_loss + b{*}obj\_loss + c{*}cls\_loss \tag{1}$$

$$a + b + c = 1 \tag{2}$$

In other words, the total loss is the weighted sum of the three losses. The weight of the confidence loss is typically given the most weight, followed by the weight of the rectangle frame loss and the weight of the categorisation loss.

The method for deriving and calculating DIoU is as follows:

$$DIoU = IoU - \left(\frac{\rho}{c}\right)^2 \tag{3}$$

$$\rho^2 = (x_p - x_1)^2 + (y_p - y_1)^2 \tag{4}$$

$$c^2 = (\max(x_{p2}, x_{l2}) - \min(x_{p1}, x_{l1}))^2 + (\max(y_{p2}, y_{l2}) - \min(y_{p1}, y_{l1}))^2 \tag{5}$$

ρ is the distance between the target frame and the centre point of the anchor frame, and c is the diagonal length of the external rectangular frame. According to the above formula, the value range of DIoU is also [−1,1]. When the two boxes coincide completely, DIoU is set to 1; when the two boxes are infinitely far apart, DIoU is set to −1.

Thus, the calculation formula of DIoU Loss can be obtained:

$$loss_{DIoU} = 1 - DIoU \tag{6}$$

In the dataset of this paper, some small citrus defects are concentrated, that is, the detection targets are relatively dense. Due to the mutual occlusion of the detection boxes, the overlapping area is very large, which is also easy to lead to missed detection and wrong detection. Therefore, the loss function GIoU was changed into DIoU in this paper. The latter is more consistent with target frame regression's mechanism than the former, and target frame regression is stabilised by taking into account the distance, rate of overlap, and scale between the target frame and anchor frame.

2.3.3. *YOLOv5 model adding attention mechanism*

The attention mechanism was first introduced into the recurrent neural network model by the Google Deep Mind team in 2014 to perform image classification, realising efficient and accurate recognition of multiple objects of images (Mnih, Heess, Graves, 2014). The basic principle of the mechanism is to achieve network adaptation through weight distribution and information filtering, as well as to find more useful data from huge feature information and send it to the convolution stack in order to train networks. In convolutional neural networks, the attention mechanism can achieve better detection results by obtaining the available attention information in the feature map. Hu, Shen, Sun (2018) proposed the Squeeze and Excitation Network (SENet) channel attention that effectively established inter-channel interdependent relations by compressing each 2D feature map. The Convolutional Block Attention Module (CBAM) proposed by Woo, Park, Lee and Kweon (2018) further advanced SENet idea by introducing spatial information encoding using large-sized convolution kernels. Based on SENet module, the Efficient Channel Attention Network (ECANet) was proposed with a local cross–channel interaction method without dimensionality reduction and an adaptive strategy to select one-dimensional convolution kernel size, thus improving effect (Wang et al., 2020). The Coordinate Attention (CA) module can be regarded as a computing unit, and its purpose is to enhance the expression capacity of the learned features in mobile network, thereby enhancing the network's ability to detect image areas (Hou, Zhou, Feng, 2021).

The lightweight attention module CBAM enables to pay attention to both the channel and spatial dimensions. CBAM consists of two independent sub-modules, Channel Attention Module (CAM) and Spatial Attention Module (SAM), which conduct Channel Attention and spatial Attention respectively. The two attention modules are linked in a sequential order. Its network structure is shown in Fig. 4. This ensures that it may be added as a plug-and-play module to the current network architecture as well as saving settings and processing resources.

The CAM Module is shown in Fig. 5. The input feature F is generated via global Max pooling and global Average pooling based on width and height, respectively, and are then delivered into a two-layer neural network. The number of neurons in the first layer is C/r (r is the reduction rate), the activation function is Relu, and the number of neurons in the second layer is C. There is a common neural network between the two levels. The final channel attention feature, M_c, is created by first combining the MLP output features based on element-wise, and then activating them with sigmoid. The input features needed by the Spatial attention module are then produced by performing element-wise multiplication between M_c and the input feature F. The calculation formula is as formula (7):

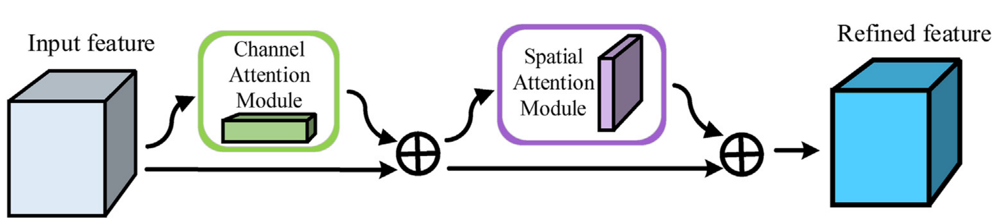


Fig. 4 – CBAM attention mechanism network structure.

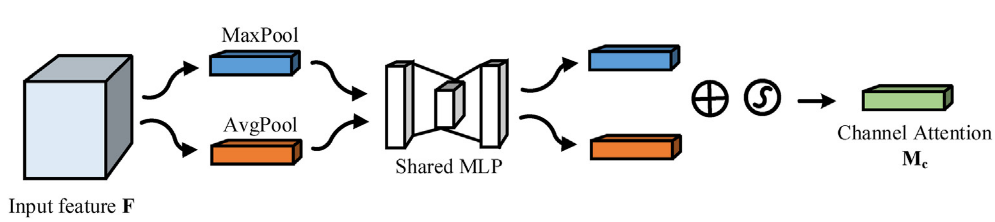


Fig. 5 – CAM network structure.

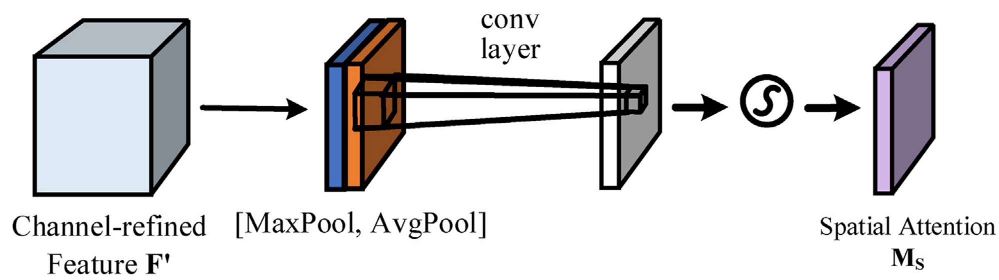


Fig. 6 – SAM network structure.

$$M_C(F) = \sigma[MLP(AvgPool(F)) + MLP(MaxPool(F))]$$

$$= \sigma\left[W_1\left(W_0\left(F^c_{avg}\right)\right) + W_1\left(W_0\left(F^c_{max}\right)\right)\right] \quad (7)$$

where W_0 and W_1 are the weights of the first and second layers respectively. σ is a sigmoid activation function, and Multilayer Perceptron (MLP) represents two fully connected shared layers. The characteristics of AvgPool and MaxPool are expressed in F^c_{avg} and F^c_{max} respectively.

The Spatial Attention Module is introduced to pay attention to which portion of the space features are important after the channel attention output. The SAM Module is shown in Fig. 6. Take the feature graph F' output by Channel Attention module as the input feature graph of this module. Firstly, conduct a channel-based global Max pooling and global Average pooling to obtain two $H \times W \times 1$ feature maps. Second, concatenation is performed on both feature maps. Then through a 7×7 convolution operation, dimension reduction is 1 channel, namely $H \times W \times 1$. The convolution kernel adopts 7×7, while keeping H and W unchanged. Finally, the Sigmoid function was used to generate the spatial weight coefficient, namely M_s. The feature is multiplied with the input feature of the module to obtain the final generated feature.

The calculation formula is presented in the following equation:

$$Ms(F) = \sigma\left[f^{7\times7}(AvgPool(F')) + f^{7\times7}(MaxPool(F'))\right]$$

$$= \sigma\left(f^{7\times7}\left(\left[F^s_{avg}; F^s_{max}\right]\right)\right) \quad (8)$$

Due to the small area of the invisible defect and the few pixel features of citrus, it is required that the inspection model can obtain more detailed information for the detection target that needs attention and suppress other useless information. In this paper, all the C3 modules in the Backbone module were replaced by the C3CBAM module that integrated into the CBAM attention mechanism. The C3CBAM module is shown in Fig. 7. Ci indicates the number of characteristic channels, and BottleNeck is formed by stacking multiple convolution

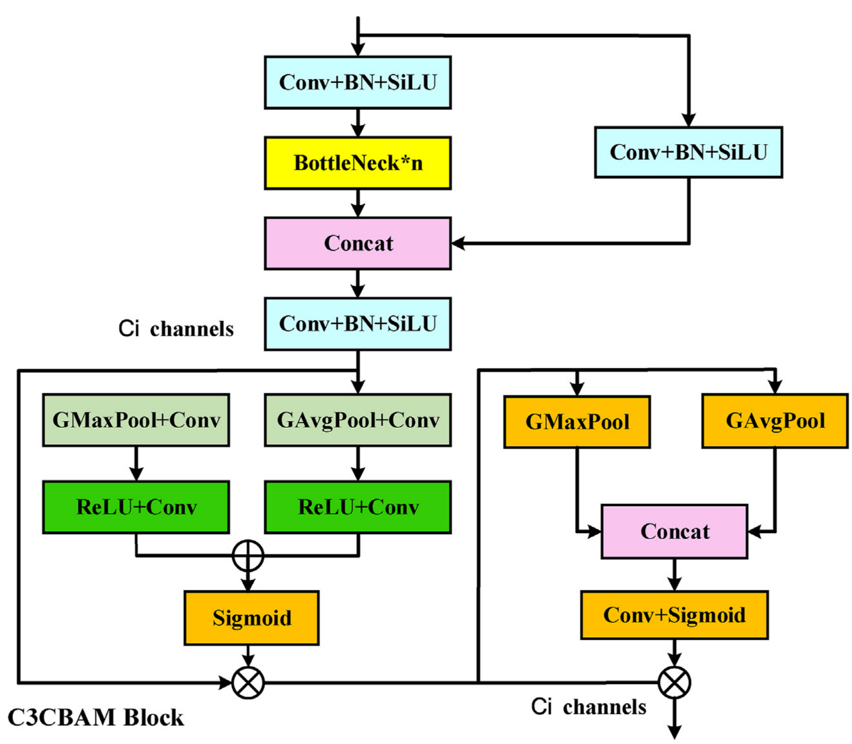


Fig. 7 – Structure diagram of C3CBAM module.

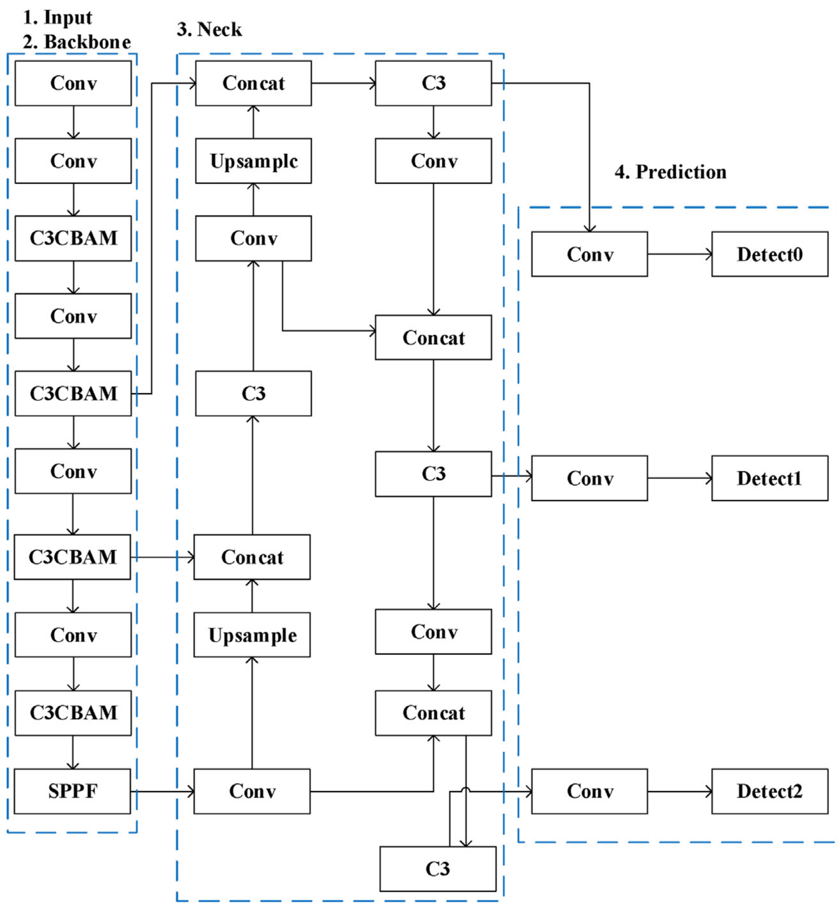


Fig. 8 – Schematic diagram of YOLOv5 network with attention mechanism.

layers. At the same time, the network could obtain wider regional information and enhance the learning ability of image features that were similar to the background information, and the network performance was enhanced at a negligible cost while avoiding an increase in the number of parameters and computations. The improved network structure is shown in Fig. 8.

2.3.4. *Experimental design*

The YOLOv5 6.0 with different depths and widths was first trained in order to improve the poor detection effects of defects with small areas under the UV fluorescent and address the challenge of separating some defects from background. Subsequently, the YOLOv5 series model with the best results was improved by changing the loss function to DIoU and adding CBAM, SE, ECA, and CA attention mechanisms, respectively, to the Backbone module for training.

Under Windows 10 and Python3.9, a training environment based on the Pytorch deep learning framework was created for the experiment, with 11.2 CUDA and 11.2 CUDNN configured in accordance with NVIDIA driver version 11.2 specifications. The CPU used in the experiment is Intel Core i7-10700 with 32G memory (RAM) at 4.59 GHz, and the GPU is GeForce RTX 3090.

The initial learning rate of the model was set to 0.01, the decay was set to 0.0005 and the momentum of the learning rate was set to 0.937. The batch size of each training was set to 8.

2.3.5. *Evaluation indicators*

The mean Average Precision (*mAP*), *Precision* and *Recall* of model training samples were used for evaluation, which defined as follows:

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

Since there are only two different types of label categories in this study, the formula reads as follows: TP is the number of correctly recognised defect classes, FP is the number of mistakenly identified backgrounds, and FN is the number of unidentified defects.

$$AP = \int_0^1 PdR \tag{11}$$

where P is the vertical coordinate Precision of the P-R curve and R is the horizontal coordinate Recall of the P-R curve.

$$mAP = \frac{1}{n}\sum\nolimits_{i=0}^{n} AP_i \tag{12}$$

where n is the total number of categories of the training sample set, and i is the number of the current category.

3. Results

3.1. YOLOv5 comparison experiments with different depths and widths

By adjusting the two parameters of model depth multiple and layer channel multiple of the YOLOv5 model, it is divided into five models of YOLOv5n, YOLOv5s, YOLOv5m, YOLOv5l and YOLOv5x. The above models have the same network architecture. Among them, YOLOv5n has the smallest model size and the number of parameters of the model, and YOLOv5x has the most.

To investigate the impact of different depths and widths of YOLOv5 on the detection of citrus defects, the comparison experiments were conducted as the preliminary experiments, the epoch was set to 300, with the results shown as Table 2.

As can be seen from Table 2, in the case of the same dataset, the YOLOv5n model has the worst detection effect, which is not suitable for the dataset with a small detection target in this paper. And with the deepening of the depth and width from YOLOv5n to YOLOv5x, the *mAP@0.5*, *Precision* and *Recall* of YOLOv5s are significantly improved compared to YOLOv5n, while from YOLOv5s to YOLOv5x, the improvement of various evaluation indicators is small, YOLOv5m and the *Precision* of YOLOv5s has decreased compared to that of YOLOv5l. At the same time, the *Precision* and *Recall* of defect1 and defect2 are gradually improved as a whole. The model's detection capacity for defect2 in particular has been greatly enhanced, but *Precision* and *Recall* have been poor since the majority of defect2 is hard to identify because it is close to the background colour information.

It can be seen that more and more residual structures and the number of convolution kernels have continuously strengthened the feature extraction and fusion capabilities of the model, and *mAP@0.5*, *Precision* and *Recall* have also been improved.

Overall, the performance of YOLOv5l and YOLOv5x are similar, but the YOLOv5x model has the best mean average precision and more advantageous *Precision* and *Recall*, indicating that the YOLOv5x model can detect more citrus defects with higher confidence and it has a higher probability of detecting citrus epidermal defects.

Table 2 – Comparison experimental results of YOLOv5 with different depths and widths.

Model Name	Defect Name	mAP@0.5	Precision	Recall
YOLOv5n	defect1	71.9	80.4	66.2
	defect2		77.4	57.3
YOLOv5s	defect1	85.3	86.0	84.1
	defect2		85.8	80.2
YOLOv5m	defect1	86.5	88.8	85.9
	defect2		81.2	82.3
YOLOv5l	defect1	89.1	89.9	87.5
	defect2		85.6	83.3
YOLOv5x	defect1	90.7	93.3	90.5
	defect2		86.7	88.5

3.2. YOLOv5x comparison experiments of different loss functions

In order to compare the performance of citrus defects detection using the YOLOv5x with different loss functions, the comparison experiments of the recognition effects were conducted before and after modifying the loss function. The epoch was set to 700. Experimental results of YOLOv5x with different loss functions on the test set are shown as Table 3.

When epoch increases from 300 to 700, YOLOv5x converges completely, the *Precision* increases by 0.4%, *mAP* decreases by 1%, and *Recall* decreases by 2%. When CIoU and EIoU were used as loss functions of YOLOv5, the results were similar. Compared with GIoU, *Recall* of both was increased by 2.1%, but *mAP* and *Precision* were decreased. When the loss function is changed to DIoU, *mAP*, *Precision* and *Recall* are increased by 0.9%, 2.2% and 2.6%, respectively. It can be seen that GIoU degrades into IoU during the convergence process, that is, when the real frame contains the prediction frame and the size of the real frame and prediction frame is fixed, the loss value remains unchanged. DIoU makes the centre point of the prediction frame close to the centre point of the real frame, and the detection effect is better for the dataset with smaller detection targets, which is more suitable for the detection of citrus surface invisibility defects.

Table 3 – Experimental results of YOLOv5x comparison with different loss functions.

Model Name	mAP@0.5	Precision	Recall
YOLOv5x(GIoU)	89.7	90.4	87.5
YOLO5x(CIoU)	87.4	86.3	89.6
YOLO5x(EIoU)	87.2	85.3	89.6
YOLOv5x(DIoU)	90.6	92.6	90.1

3.3. Comparative experiments of fusion attention mechanism

Comparative experiments were conducted to analyse the performance influence of the CBAM, SE, ECA, and CA modules on the entire network in order to assess the improved detection model of citrus epidermal defects. Set the epoch to 700. The loss function in YOLOv5x was changed to DIoU and named YOLOv5x'. The best weight *best.pt* trained by YOLOv5x′ was used as the weights for the comparative experiments to increase model training accuracy and convergence speed. The results of the comparative experiments are shown as Table 4.

Before the YOLOv5x network was improved, the most critical problem was the poor detection effect of defect2, which was similar to the background colour information. After replacing all C3 modules with C3 module incorporating attention mechanism, the *AP* of C3CBAM for defect1 improved, while that of C3CA, C3SE and C3ECA decreased slightly, among which C3ECA decreased the most. For defect2, the detection effects of C3CBAM, C3SE and C3CA modules all improved, while C3ECA decreased, and the growth rates were

Table 4 – Results of comparative experiments with fused attention mechanisms.

Model Name	Defect Name	AP	Precision	Recall
YOLOv5x′	defect1	93.9	92.6	90.1
	defect2	87.4		
YOLOv5x′-C3CA	defect1	93.0	90.9	91.3
	defect2	90.8		
YOLOv5x′-C3CBAM	defect1	97.4	94.0	95.1
	defect2	93.5		
YOLOv5x′-C3ECA	defect1	85.9	90.0	86.1
	defect2	84.4		
YOLOv5x′-C3SE	defect1	92.1	91.4	93.4
	defect2	90.0		

Table 5 – Results of ablation experiments.

Model Name	mAP@0.5	Precision	Recall	Average detection speed (ms per pic)
YOLOv5x	89.7	90.4	87.5	66.5
YOLOv5x′	90.6	92.6	90.1	
YOLOv5x-CBAM	91.4	93.3	91.0	
YOLOv5x-C3CBAM	91.4	93.6	91.9	
YOLOv5x′-C3CBAM	95.5	94.0	95.1	44.4

successively C3CBAM, C3CA and C3SE. Compared with YOLOv5x’, it could be seen that the *Precision* improved the most when C3CBAM module was introduced, followed by C3SE, C3CA and C3ECA, which all decrease to a certain extent

Table 6 – Results of comparative experiments.

Model Name	Defect Name	AP	mAP@0.5
YOLOv5x	defect1	93.6	89.7
	defect2	85.8	
YOLOv5x′-C3CBAM	defect1	97.4	95.5
	defect2	93.5	
Faster R–CNN based on Detectron2	defect1	43.0	37.5
	defect2	31.9	

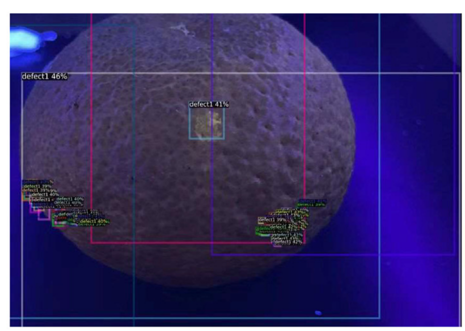
in the order from large to small. *Recall* improved the most when C3CBAM and C3SE modules were introduced, followed by C3CA and a decrease in C3ECA. The above results fully indicate that C3CBAM module can more fully learn the features of defect2 defect area and distinguish it from the background, so as to improve the detection accuracy of citrus epidermal invisible defects.

Therefore, by introducing the attention CBAM module, the location information and channel information can be better extracted to ensure that the more valuable channel features and location features occupy a higher proportion in the feature map. Overall, the most significant improvement was achieved when the CBAM module was introduced, and the detection of citrus epidermis defects was improved.

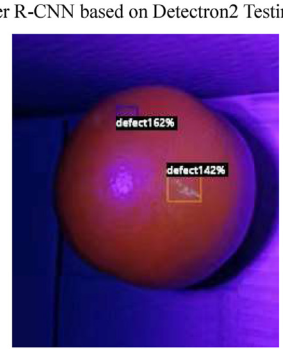
3.4. *Ablation experiments*

In order to verify the effectiveness of the improved model based on YOLOv5x, ablation experiments were performed. Set the epoch to 700. The results of the detection on the test set are shown in Table 5.

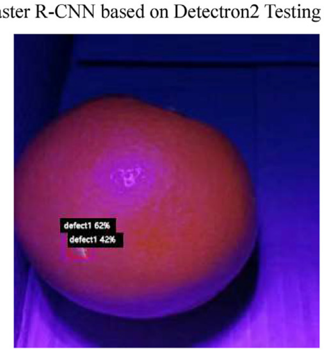
a. Faster R-CNN based on Detectron2 Testing effect



b. Faster R-CNN based on Detectron2 Testing effect

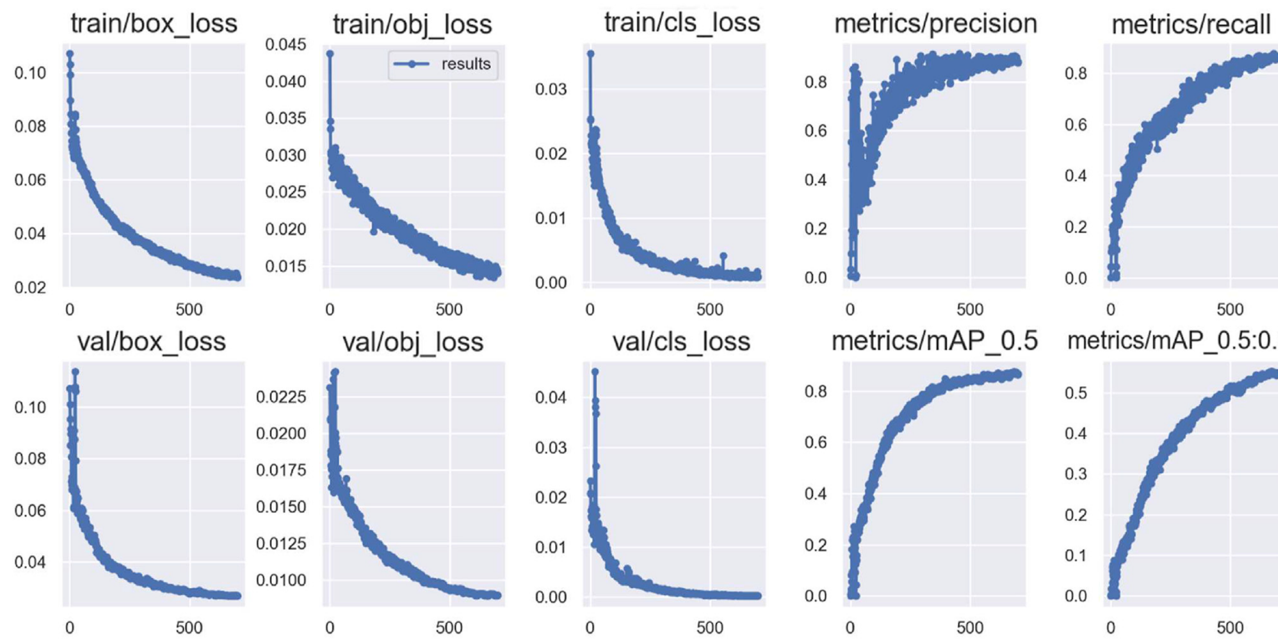


c. Faster R-CNN based on Detectron2 Testing effect

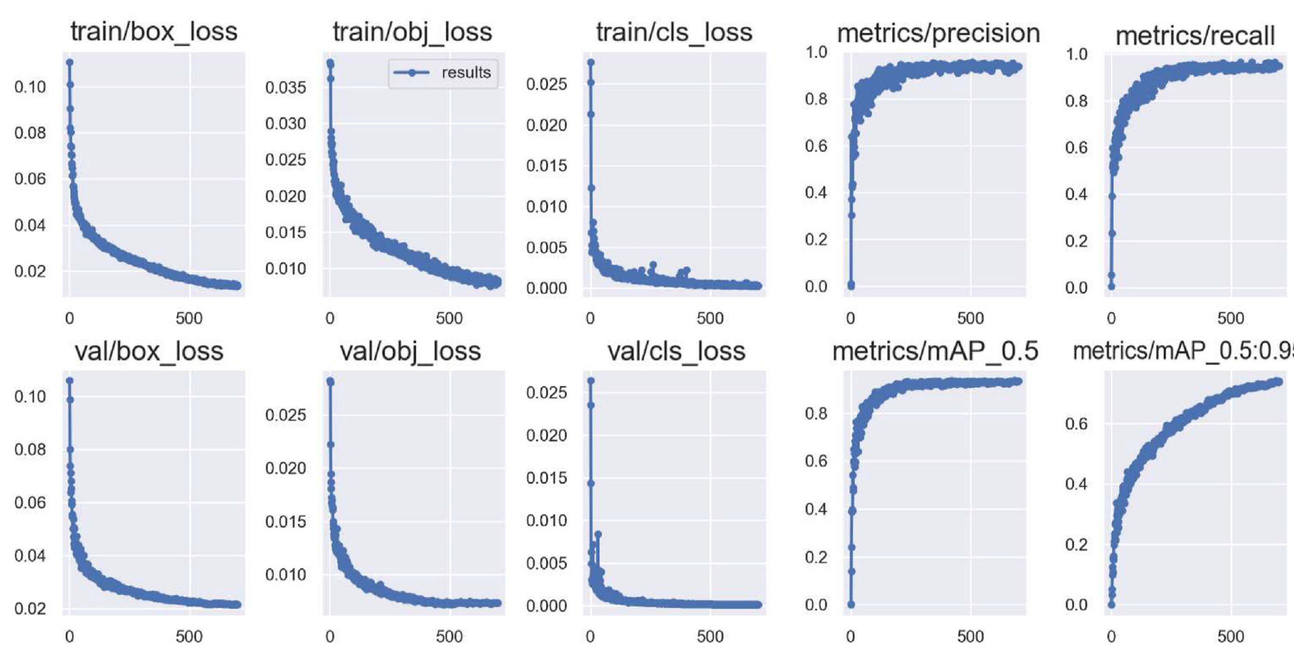


d. Faster R-CNN based on Detectron2 Testing effect

Fig. 9 – The results of Faster R–CNN based on Detectron2 to detect citrus defects.



a. YOLOv5x



b. YOLOv5x'

Fig. 10 – Training results of YOLOv5x and YOLOv5x'. (a) YOLOv5x, (b) YOLOv5x′.

When the attention module CBAM is fused in the YOLOv5x model, the YOLOv5x-CBAM obtained by directly adding the CBAM module before the last layer of SPPF of the backbone structure is compared with YOLOv5x, *mAP@0.5* increased by 1.7%, *Precision* increased by 2.9%, *Recall* increased by 3.5%, indicating that the addition of attention module had a positive effect on the detection effect of the model. Compared with YOLOv5x-C3CBAM, the *Precision* and *Recall* of YOLOv5x -C3CBAM are increased by 0.3% and 0.9% respectively, which indicates that replacing the original C3 module with CBAM is more effective than directly adding a layer of CBAM structure. Compared with YOLOv5x '-C3CBAM using the same loss function regression, *mAP@0.5*, *Recall* and *Precision* are increased by 4.9%, 1.4% and 5%, respectively, suggesting that

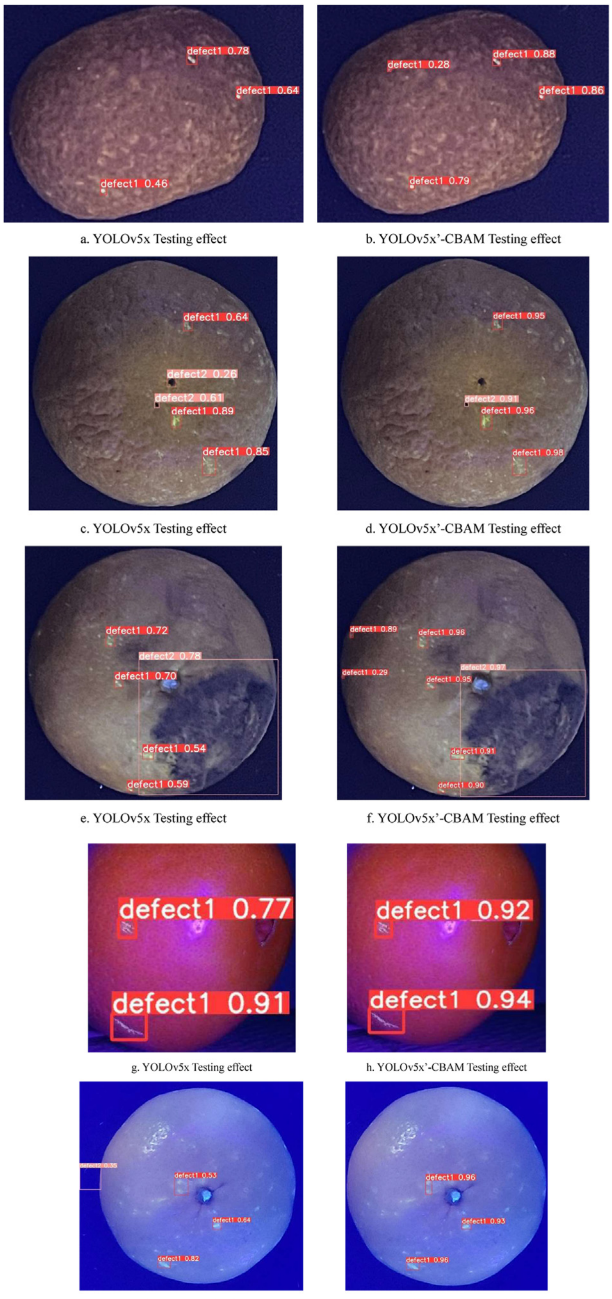


Fig. 11 – Comparison of detection effect before and after YOLOv5 improvement.

the network integrated with attention mechanism can greatly improve the detection performance of invisible defects in citrus epidermis. Finally, compared with *mAP@0.5* of YOLOv5x, the improved YOLOv5x '-C3CBAM improves by 5.8%, *Precision* by 3.6%, *Recall* by 7.6%, and average detection speed by 22.1 ms per pic. Therefore, the above results can fully illustrate the effect of this paper on the model improvement.

3.5. *Experiments of contrast*

In order to compare the effectiveness and performance of the one-stage target detection model with the common two-stage model, such as the improved model based on YOLOv5x and the Faster R–CNN model based on Detectron2, the comparative experiments were carried out. Set the epoch to 700. The initial learning rate of the model was set to 0.01, the decay was set to 0.0005 and the momentum of the learning rate was set to 0.937. The batch size of each training was set to 8. The detection results of the test set are shown in Table 6.

As can be seen from Table 6, Faster R–CNN based on Detectron2 amplified the detection difficulty of the invisible defect detection dataset of citrus in this paper. In contrast, the detection performance of YOLOv5x has advantages. In addition, we used Faster R–CNN based on Detectron2 to detect citrus defects, and the results are shown in Fig. 9.

As can be seen from Fig. 9, this model has a very serious situation of repeated detection and misclassification of defect types in citrus defect detection, and even almost completely fails to distinguish defect2 with similar background colour information. Secondly, because Faster R–CNN based on Detectron2 did not have NMS suppression when predicting, a large number of detection boxes would appear.

3.6. *Result analysis*

The training process of YOLOv5x and YOLOv5x′ before and after modifying the loss function is shown in Fig. 10. The test set is entered into the model for testing when the model training is finished. The identification and comparison results of YOLOv5x and YOLOv5x′-CBAM before and after the improvement of the model are shown in Fig. 11 at the same confidence threshold.

As shown in Fig. 10, The value of the loss function decreases in the training process. The stochastic gradient descent algorithm is used to optimise the network, and the network weight and other parameters are constantly updated. Before the training batch reached 500, the loss function value decreased rapidly, and the accuracy rate, recall rate and average accuracy rate increased rapidly. The network continues to iterate. When the training batch reached about 500, the decline of the loss function value gradually slowed down. Similarly, the increase in parameters such as average accuracy has slowed down. When the training batch reached 500, the loss curve of the training set and the validation set showed a little downward trend, and the other index values also tended to be stable. The network model basically reaches the convergence state, and the optimal network weight is obtained at the end of training. Comparing a and b in Fig. 10, it can be seen that the values of train/box_loss and val/box_loss of YOLOv5x are both slightly higher than those of YOLOv5x ‘when epoch is 0, which may be because the distance between the original anchor frame and the target frame is slightly larger. When epoch is 0, the values of train/cls_loss and val/cls_loss of YOLOv5x are both slightly larger than those of YOLOv5x ', indicating that the classification of YOLOv5x ‘algorithm is more accurate. The precision of YOLOv5x ‘is different from that of YOLOv5x′, which increases first and then decreases, indicating that YOLOv5x ‘converges faster after modifying the loss function. In fact, YOLOv5x′ almost converges faster than YOLOv5x '. It shows that DIoU enables the algorithm to better locate the centre point of the detection target.

Table 7 – Some researches on fruit defect detection in recent years.

Object	Network	Defect condition	Dataset condition		Network performance				
			Image content	Image background	*Recall*	*Accuracy*	*F1 score*	*Precision*	*mAP*
Defects on tomatoes (Da Costa, Figueroa, Fracarolli, 2020)	ResNet	Longitudinal cut, dark spot, rain bruise and deformity	A damaged tomato	Laboratory	86.6%			91.7%	94.6%
Green plum defects (Zhou, Zhuang, Liu, Liu, 2020)	Based on the improved VGG network	Rot, spots, scars, cracks	A damaged green plum	Laboratory		93.8%			
Defective citrus (Kukreja and Dhiman, 2020)	Dense CNN	Good and damaged	A defective citrus	Laboratory and orchard		89.1%			
Apple skin defects (Valdez, 2020)	Based on the improved YOLOv3	Good and damaged	Multiple defective apples	Orchard					74.3%
Defective carrots (Xie, Wei, Zheng, Yang, 2021)	CarrotNet based on DCNN	Fractures, deformities, bruises, split ends	A damaged carrot	Laboratory		97.04%			
Defective citrus (Chen, An, Gao, Li, 2021)	Mobile-Citrus consists of MobileNet-V2 and PANet	Mechanical damage and skin lesions	Multiple defective citrus	Laboratory	87.0%	88.0%	87.1%		
Defective citrus (Zhang, Xun, Chen, 2022)	YOLOv4 and EfficientNet	Canker, anthracnose, sunscald, greening, and melanose	Multiple defective citrus	Orchard		89.0%	87.2%		
Apple skin defects (Fan et al., 2022)	Based on the improved YOLOv4		Multiple defective apples	Laboratory					93.74%
Citrus epidermal defects (this paper)	Based on the improved YOLOv5	Injury and scar	A defective citrus	Laboratory	95.1%			94.0%	95.5%

As shown in Fig. 11, the overall effect of the improved YOLOv5x model seems to be better than the YOLOv5x model, and more citrus epidermal defects can be detected with higher accuracy. The improved network can more accurately detect defect2 that is close to the background colour information and improve the situation of false and missed detection.

In summary, the improved model based on YOLOv5x in this paper can more accurately identify the location and category of defects and improve the accuracy of defects detection. The model is not only suitable for detecting invisible defects of citrus epidermis, but also able to solve the problem of difficult detection of invisible defects under natural light to a certain extent.

4. Discussion

In this study, we proposed a citrus invisible defect detection technique based on the improved YOLOv5. The dataset consists of citrus images containing damage, scars, etc., that is, the colour and texture information of the defect area is highly similar to that of the normal area under natural light, and the invisible defects are clearly visible after excitation by ultraviolet light. Although the fluorescence response of citrus has attracted wide attention of scholars, and the detection of citrus invisible defects has been studied based on this, it is more difficult to collect the dataset of citrus invisible defects, and there are few similar mass images used as deep learning dataset in the past. In addition, the collection of this dataset relies on the dual illumination image acquisition system designed in this paper. The detection of invisible defects based on fluorescence reactions is more convincing and verifiable since controlled experiments with white and ultraviolet light contrast are performed at the beginning of these images.

We have studied recent studies on fruit defect detection, which are organized into tables as shown in Table 7. In recent years, most of the work on fruit defect detection is based on convolutional neural network. The dataset used in the literature mainly consists of an image with a single object in the laboratory environment as the dataset, and a dataset with multiple objects in the orchard environment as the background. However, the dataset used in the above studies are different, so the performance of the detection networks used cannot be directly compared. For fair evaluation, the image content and background, network and detection performance of the dataset are selected for comparative study. As can be seen from Table 7, the detection effect of the optimized detection network is better than that of the original network, and the detection difficulty of the dataset collected in the orchard environment is usually greater than that in the laboratory environment. This indicates that it is effective and necessary to improve the network when the image contains multiple target objects and complex backgrounds.

Therefore, in this study, the analysis of the results obtained from the experiments on the YOLOv5 model with different depths and widths found that the difficulty in the detection of the dataset in this paper lies in the identification of defects close to the background information. Then, this paper optimizes the YOLOv5 model by combining the attention mechanism CBAM and modifying the loss function to DIoU, and the performance of the improved model is verified by comparison and ablation experiments. The experimental results show that the improved network can detect the invisible defects of citrus more accurately and faster from the complex background. In other words, the improved network proposed by us has better detection performance and better inference speed for citrus invisible defects, which will be conducive to deployment on edge devices. In future studies, firstly, the model will be extended to more varieties and defect types of citrus, and secondly, the model will be improved into a more lightweight network, so as to further improve the robustness and real-time performance of the detection algorithms for citrus epidermal defects.

5. Conclusion

Intelligent detection of postharvest citrus quality is of great significance to production and value assurance. Detection of citrus epidermis defect is one of the main links in postharvest citrus process. In the process of citrus epidermal defects detection, there are problems such as uneven illumination of fruits, difficulty in identifying invisible defects under natural light, and colour information of defects being similar to the background, resulting in a low detection accuracy. In order to improve the accuracy of intelligent detection of postharvest citrus quality, it is considered that the “invisible defects” of citrus epidermis are difficult to identify under natural light, but easy to identify under ultraviolet light. Therefore, in order to effectively highlight the biological characteristics of citrus epidermis, in this paper, the wavelength of UV lamp with the largest fluorescence reaction intensity was determined by experiment, and Zhejiang Linhai Citrus with the strongest fluorescence reaction was selected as the experimental object. A dual-lamp image acquisition system was designed to obtain images of citrus under halogen and 356 nm UV lamps, respectively. Then, the loss function GIoU of YOLOv5 model was changed to DIoU to optimize the problem that the loss value of YOLOv5 model was unchanged when the real box contained the prediction box and the size of the two was fixed, and the network convergence speed was accelerated. Meanwhile, the attention CBAM module was integrated into the backbone of YOLOv5 model, and the original C3 structure was replaced by C3CBAM module. The model's sensitivity to the defect information in citrus photos was increased, which also improved the YOLOv5 model's ability to detect defects that were close to the background data. Finally, the invisible defects of citrus epidermis were detected and classified based on the improved YOLOv5x model.

Deep learning holds the key to the intelligent detection of citrus defects. The fusion of fluorescence image technology and deep learning is applied in the citrus invisible defect detection proposed in this paper. We propose a detection technology for invisible defects of citrus epidermis based on an improved YOLOv5, and the effectiveness and feasibility of the proposed method are verified by experiments. The results showed that the *AP* of citrus epidermis defects detected by the improved YOLOv5 model was 0.974, the *AP* of citrus epidermis black spots was 0.935, and the *mAP* was 0.955. Compared with the YOLOv5x, which worked best for the dataset in this paper among the series of YOLOv5 models, the *mAP* of the improved YOLOv5 model was increased by 5.8%, the *Precision* increased by 3.6% and *Recall* increased by 7.6%, and average detection speed increased by 22.1 ms per pic. The proposed method effectively improves the detection performance of invisible defects in citrus epidermis, and provides technical reference for intelligent detection of citrus quality after harvest. It can also be used as a reference for defect detection in other fruits such as tomatoes and kiwi fruit.

All authors disclosed no relevant relationships.

Author contributions

Wen-Xin Hu and Jun-Tao Xiong contributed to the design of data collection, experimental design, completion of experiments, interpretation of results, and preparation of the manuscript. Jun-Hao Liang and Zhi-Ming Xie contributed to the experimental work. Zhi-Yu Liu, Qi-Yin Huang and Zhen-Gang Yang contributed to the writing of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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